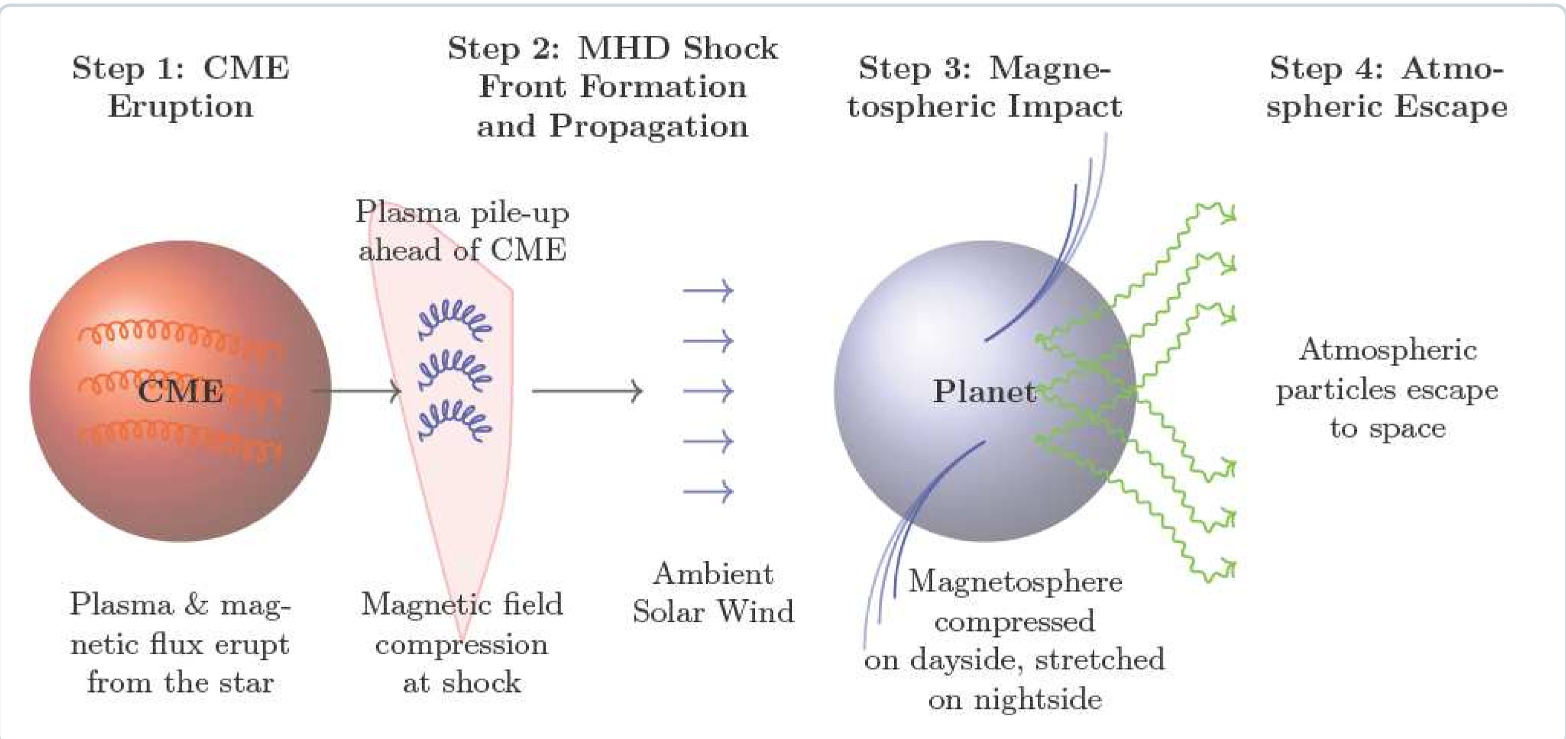
Surviving Stellar Fury: Atmospheric Escape caused by Interplanetary Shocks

INTRO

BACKGROUND

- Habitability depends on far more than simply being in the habitable zone. Without an atmosphere, liquid water stability, climate regulation, and biological protection become impossible. An exoplanet's viability depends completely on the **stability of its atmosphere**.
- Space weather conditions like solar winds, energetic particle events, and Coronal Mass Ejections (CMEs) constantly interact with planetary magnetospheres, risking stripping away atmospheres entirely over millions of years.
- **Interplanetary Shocks (IPS)** form when fast CME winds collide with surrounding slow solar winds. They feature immense density, pressure, and magnetic field amplifications that severely deform magnetospheres and erode atmospheric envelopes via ion pickup and thermal escape.



PAST STUDIES

- Previous research highlights that solar winds induce stripping of light molecules (e.g., hydrogen), altering the composition and debilitating planetary greenhouse balances (Airapetian et al., 2020).
- While solar wind effects are widely studied holistically, research on Interplanetary Shocks has primarily been restricted computationally to Earth and Mars.
- Recent studies indicate IPS severely deforms magnetic fields and accelerates ion escape (Oliveira, 2023). However, these models predominantly neglect exoplanetary diversity in mass, volume, and magnetic strength, leaving active sun-like star system habitability highly uncertain.

RESEARCH QUESTION

QUESTION

What percentage of the atmosphere in various exoplanets will be lost due to interplanetary shocks caused by CMEs from their host star over a threshold of 1,000,000 years, and how do structural morphologies dictate long-term habitability?

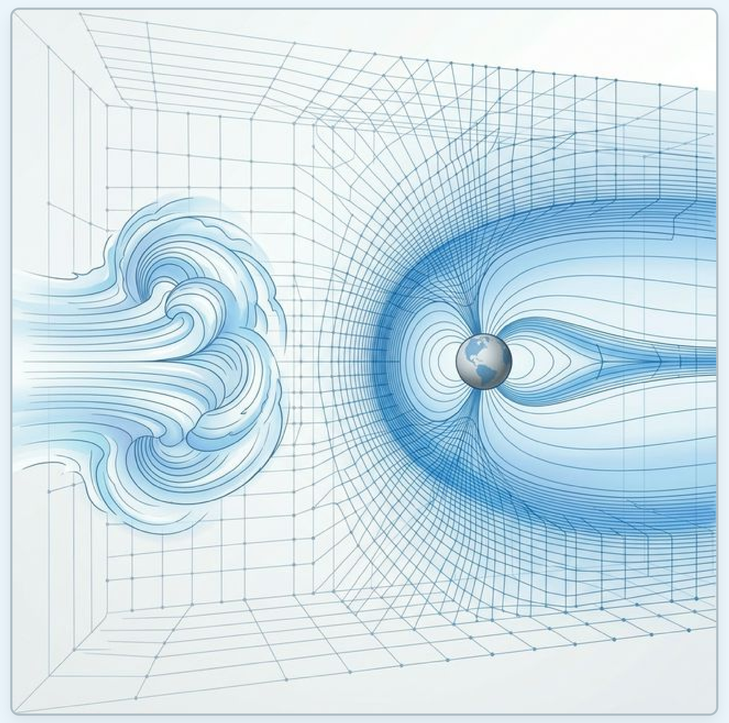
HYPOTHESIS

It is hypothesized that exoplanets with weaker magnetospheres, closer orbits, and lighter atmospheric particles lose a greater percentage of their atmospheres compared to planets with stronger magnetic fields and distant orbits. Shocks carrying highly charged plasma interact vigorously with smaller magnetospheres, allowing for energetic particle penetration, ionospheric heating, and aggressive hydrodynamic atmospheric stripping via thermal expansion and sputtering mechanisms.

METHOD

Independent Variables: Orbital distance (Semi-Major Axis 0.027 AU to 0.098 AU), Exoplanetary Radius, Mass, and scaled Stellar Wind Conditions.

Targets: A broad cross-section including Hot Neptunes (TOI-5398b), Hot Jupiters (WASP-26b, WASP-5b), and Highly-Inflated Saturns (WASP-39b). An Earth-identical baseline absent of CMEs with a standardized magnetic dipole 3.11×10^{-5} T was applied universally across all models.



Multi-stage MHD pipeline Architecture:

- **Part A (Solar Wind & CME):** Uses SWMF in spherical dimensions out to 20 solar radii with 256 × 128 × 256 grid space. Employs the NASA SolarSoft PFSS model to synthesize a G-type star's 150 km/s solar winds and randomly triggered Gibson-Low CMEs.
- **Part B (Interplanetary Space):** Propagates shockwaves utilizing Constrained Transport (CT), HLLD capturing, and MUSCL reconstruction with SSP-RK3 temporal integration. Adaptive Mesh Refinement (AMR) is finely modulated by an autonomous shock sensor tracking immense Mach number discontinuities.
- **Part C (Planetary Interaction):** Integrates PLUTO algorithms within a Cartesian computing box, utilizing 32^3 AMR block sizing. Accurately maps dynamic shock pressures driving magnetopause compression. Resolves boundary field-aligned currents directly translated via a Ridley Serial solver to model atmospheric hydrodynamic escape based on particle deposit.

RESULTS

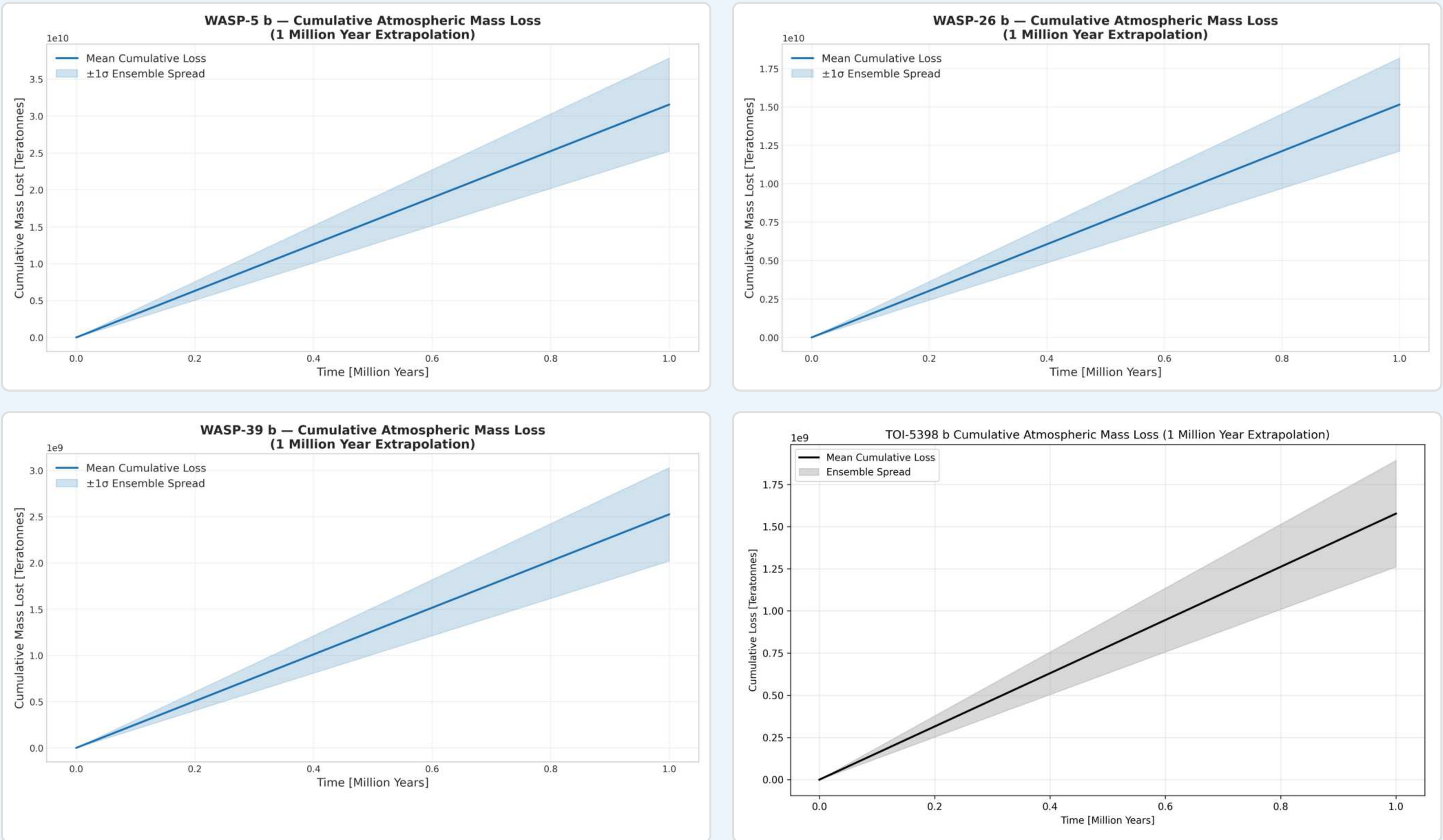


Figure 1: Cumulative Mass Loss Forecasts over 1 Myr (WASP-5b, WASP-26b, WASP-39b, TOI-5398b) showing 200 variance trials representing CME impacts with spread limits (±1σ).

RESULTS (CONT.)

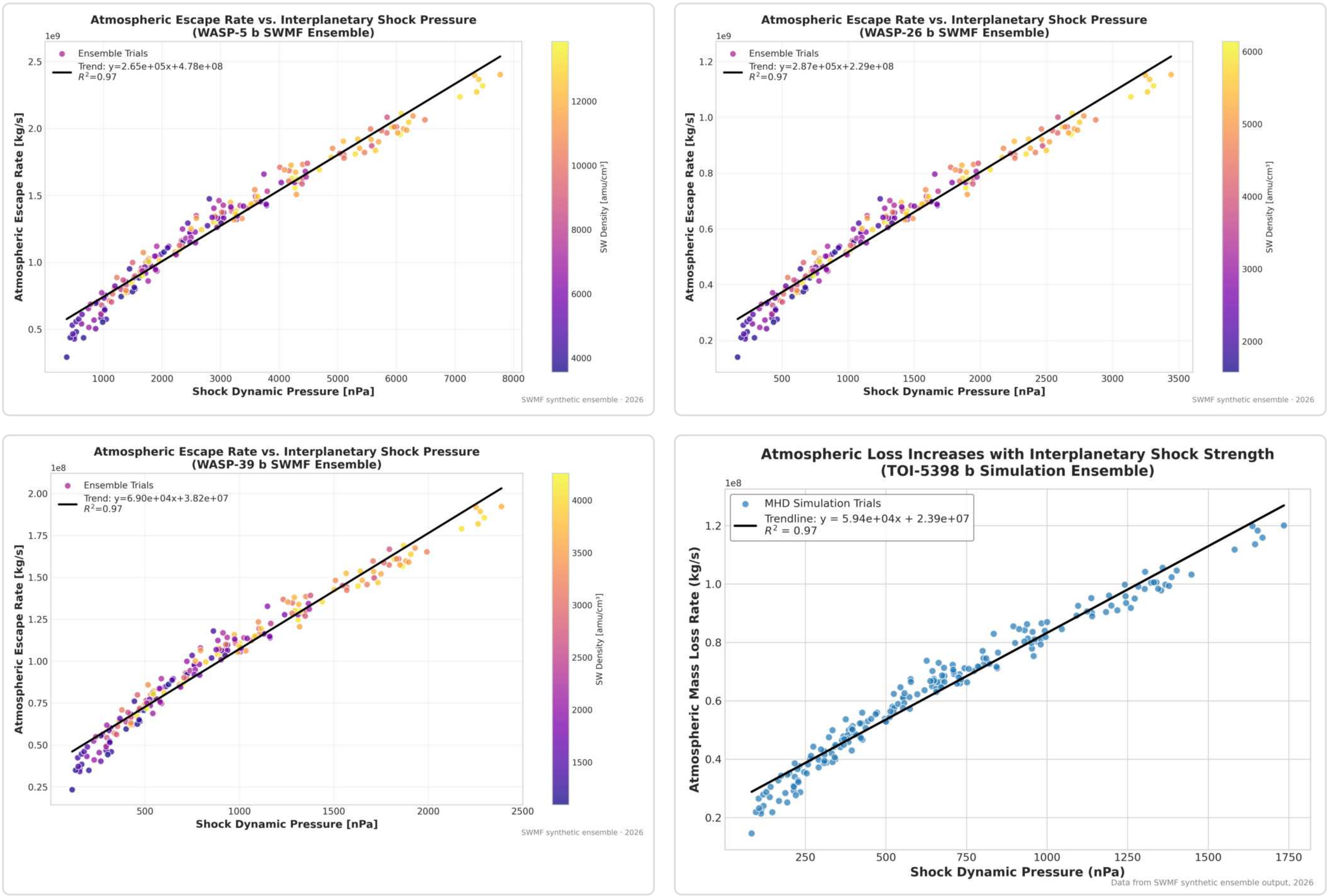


Figure 2: Super-Linear scaling showing Atmospheric Escape magnitude dependent on Interplanetary Shock Pressure. ANOVA test proved targets were astronomical deviations (F = 879.4, p < 1e-252).

ANALYSIS

Proximity Exponentially Dominates Loss: Orbital separation aggressively dictates catastrophic atmospheric stripping. WASP-5b (0.027 AU orbit) endures mass escape forces spanning 100% greater than WASP-26b (0.040 AU), generating a mean escape rate of 1.26×10^9 kg/s.

Power-Law Correlation: Computation validates a fundamental super-linear power-law function ($L \propto P_{dyn}^{1.19}$) mapping shock dynamic pressure directly to atmospheric escape.

Puffy Planets Secured By Distance: Inflation proves less vital than spatial distance: WASP-39b's severely inflated, low-mass morphology maintained substantially reduced stripping proportions strictly due to a slight orbit expansion (0.048 AU) which prevented intense direct CME interaction.

CONCLUSION

Interplanetary shocks definitively manifest as a catastrophic driving mechanism controlling exoplanet atmospheric resilience. Reoccurring extreme CME weather rapidly collapses biosphere viability exponentially faster when environments cross ~0.03 AU thresholds. Magnetic shielding constraints, orbital proximity arrays, and rigorous space weather evaluations remain non-negotiable considerations for accurate planetary habitability categorizations throughout the cosmos.

FUTURE WORK

Future research paradigms will investigate expanding algorithmic boundaries toward **M-dwarf systems** and tidally locked exoplanetary morphologies. Progressive developments plan to incorporate complex **Particle-in-cell (PIC) coupling** frameworks alongside full ionosphere chemistry models into natively unified machine learning acceleration engines to resolve chaotic multidimensional plasma dynamics.

REFERENCES

Constraints: unobservable empirical magnetic fields required exact Earth-like dipole approximations combined with simplified environmental chemistry limitations.

Airapetian, V. S., et al. (2020). Impact of Space Weather on Climate and Habitability. *Int. Journal of Astrobiology*, 19(2), 136-194.

Oliveira, D. M. (2023). Geoeffectiveness of interplanetary shocks. *Frontiers in Astronomy*, 10.

Varela, J., et al. (2023). Earth's habitability over the Sun's main-sequence history. *MNRAS*.